

Databricks Lab : Workspace Directory & Notebook Setup

Duration : 60 Min.

Scenario

As a Junior Data Engineer at LabsKraft, you are tasked with provisioning a shared development space and template notebook for your data science team. Your objective is to create a workspace directory and initialize a Python template notebook with an interactive greeting script.

Input Details

The environment has been pre-configured with the following resources:

- **Workspace:** Pre-configured through KodeBuck Service credentials (`DATABRICKS_HOST`, `DATABRICKS_TOKEN`)
- **Resource Group:** `rg-iRUN-LTM-Assessment`

Username & Naming Conventions

Your candidate prefix is a combination of your username (the part of your email/login before the @ or _ symbol) and exam code. For example, if your username is student (or `student_labskraft.com`) and your exam code is exam123, your prefix is **student-exam123**.

You must name your resources accordingly:

- **Workspace Directory Path:** `/Shared/<prefix>-workspace` (e.g. `/Shared/student-exam123-workspace`)
- **Notebook Name:** `data-analysis`

Task Objectives

Perform the following configuration steps directly in the Databricks Workspace Console or using the Databricks API:

1. Create Workspace Directory & Notebook

- **Create Workspace Directory:** Create a directory named `/Shared/<prefix>-workspace` under the Shared folder (e.g. `/Shared/student-exam123-workspace`).
- **Create Template Notebook:** Inside the directory `/Shared/<prefix>-workspace`, create a notebook named `data-analysis`.
- **Configure Notebook Language:** Set the default language of the notebook to **Python**.
- **Initialize Interactive Code:** Write a Python script in the notebook that:
 - Prompts the user to enter their name using the `input()` function.
 - Prints the welcome message: `Welcome to KodeBuck Databricks Workspace, <name>!` (where `<name>` is the dynamic value supplied via input).

Example snippet for the notebook content:

```
name = input("Enter your name: ")
print(f"Welcome to KodeBuck Databricks Workspace, {name}!")
```

Verification

Once you have performed the tasks, you can run the verification script to check your progress and receive your score. The verification system will check the actual resources in the Databricks Workspace.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Directory existence (/Shared/<prefix>-workspace exists)	4 Marks
TC2	Notebook existence (/Shared/<prefix>-workspace/data-analysis exists)	4 Marks
TC3	Object Type verification (path resolves to a notebook object)	4 Marks
TC4	Notebook Language (language is PYTHON)	4 Marks
TC5	Interactive Code Check (uses input () and prints "Welcome to KodeBuck Databricks Workspace")	4 Marks

Total Score: 20 Marks

Important Notes

- Ensure all resource names contain your candidate username prefix in lowercase.
- The notebook path must exactly match the naming convention.